

ANSWER KEY — Integration Lesson 26 Test: u-Substitution

Mastery

For the grader · full working shown · give credit for correct method with small arithmetic slips · limits plugged into the wrong variable's formula is THE error to watch for

1. Compute $\int_0^1 2x(x^2+2)^2 dx$ with changed limits.

Work: $u = x^2+2$, $du = 2x dx$ (already present). Limits: $u(0) = 2$, $u(1) = 3$. Integral = $\int_2^3 u^2 du = [u^3/3]_2^3 = 27/3 - 8/3 = 19/3$. **Answer: 19/3.** (Grader: back-substituting to $(x^2+2)^3/3$ with x-limits 0,1 $\rightarrow 27/3 - 8/3 = 19/3$ also earns full credit, though the prompt asked for changed limits — take off half if limits were never converted.)

2. Find the new u-limits: (a) $u = x^3+1$, x : 0 to 2. (b) $u = \cos x$, x : 0 to $\pi/2$.

Work: plug the x-limits into the u formula. (a) $u(0) = 0+1 = 1$; $u(2) = 8+1 = 9 \rightarrow u$: 1 $\rightarrow$ 9. (b) $u(0) = \cos 0 = 1$; $u(\pi/2) = 0 \rightarrow u$: 1 $\rightarrow$ 0. **Answer: (a) 1 $\rightarrow$ 9 (b) 1 $\rightarrow$ 0.** (Grader: (b) running downhill is CORRECT — "reversed" limits are legal; don't accept $0 \rightarrow 1$ unless the student explains they flipped the integral's sign.)

3. Find $\int x/(x^2+9) dx$.

Work: derivative of bottom is $2x$; top is x — off by constant 2, so borrow it: $u = x^2+9$, $x dx = du/2$. $(1/2)\int du/u = (1/2)\ln|u| = (1/2)\ln(x^2+9)$ (bottom always positive). Check: derivative = $(1/2) \cdot 2x/(x^2+9) = x/(x^2+9)$ ✓. **Answer: $(1/2) \ln(x^2+9) + C$.** (Grader: arctan-form answer $(1/3)\arctan(x/3)$ is wrong — that's for $1/(9+x^2)$ with NO x on top.)

4. Compute $\int_1^{e^2} (\ln x)/x dx$ with changed limits.

Work: $u = \ln x$, $du = (1/x)dx$. Limits: $u(1) = \ln 1 = 0$; $u(e^2) = \ln(e^2) = 2$. Integral = $\int_0^2 u du = [u^2/2]_0^2 = 4/2 - 0 = 2$. **Answer: 2.**

5. Find $\int \cos^4 x \sin x dx$.

Work: $u = \cos x$, $du = -\sin x dx \rightarrow \sin x dx = -du$. Integral = $-\int u^4 du = -u^5/5 = -\cos^5 x/5$. Check: $d/dx[-\cos^5 x/5] = -(5\cos^4 x)(-\sin x)/5 = \cos^4 x \sin x$ ✓. **Answer: $-\cos^5 x/5 + C$.** (Grader: the missing minus sign is the expected slip; $+\cos^5 x/5$ gets no more than half.)

6. Compute $\int_0^{\pi/2} \sin^4 x \cos x dx$ with changed limits.

Work: $u = \sin x$, $du = \cos x dx$. Limits: $u(0) = 0$, $u(\pi/2) = 1$. Integral = $\int_0^1 u^4 du = [u^5/5]_0^1 = 1/5$. **Answer: 1/5.**

7. Find $\int e^x/(e^x+5) dx$ and name the pattern.

Work: derivative of the bottom (e^x+5) is e^x = the top, so this is the $\int f'/f = \ln|f|$ pattern ("derivative-of-bottom-on-top $\rightarrow \ln$ "). $u = e^x+5$, $du = e^x dx \rightarrow \int du/u = \ln(e^x+5)$ (always positive, no $|\cdot|$). **Answer: $\ln(e^x+5) + C$; pattern: $\int f'/f = \ln|f|$.**

8. Compute $\int_0^{\pi/4} \sec^2 x/(1+\tan x) dx$ exactly.

Work: derivative of the bottom ($1+\tan x$) is $\sec^2 x$ = the top $\rightarrow f'/f$ pattern. $u = 1+\tan x$, $du = \sec^2 x dx$. Limits: $u(0) = 1+\tan 0 = 1$; $u(\pi/4) = 1+\tan(\pi/4) = 1+1 = 2$. Integral = $\int_1^2 du/u = [\ln u]_1^2 = \ln 2 - \ln 1 = \ln 2$. **Answer: $\ln 2$.** (Grader: this is an AP classic; full marks require the changed limits $1 \rightarrow 2$ or a correct back-substitution with x-limits.)

9. Student writes $\int_0^1 x(x^2+1)^2 dx = (1/2)\int_0^1 u^2 du = 1/6$. Find the error; give the correct answer.

Work: the error: the student kept the x-limits 0 and 1 on a u-integral — limits belong to the variable next to the d, so they had to be converted: $u(0) = 1$, $u(1) = 2$. Correct: $(1/2)\int_1^2 u^2 du = (1/2)(8/3 - 1/3) = (1/2)(7/3) = 7/6$. **Answer: forgot to change the limits; correct value 7/6.** (Grader: naming the error without recomputing = half; recomputing without naming the error = half.)

10. Compute $\int_0^{\pi/4} \tan^3 x \sec^2 x dx$ with changed limits.

Work: $u = \tan x$, $du = \sec^2 x dx$ (sitting right there). Limits: $u(0) = \tan 0 = 0$; $u(\pi/4) = \tan(\pi/4) = 1$. Integral = $\int_0^1 u^3 du = [u^4/4]_0^1 = 1/4$. **Answer: $1/4$.** (Grader: the $u = \sec x$ road also works — $\sec^4 x/4 - \sec^2 x/2$ evaluated... but it's longer; any correct route earning $1/4$ gets the point.)